Exercises IV : Topology III

October 16, 2025

---

1. Recall that S^2 is the 2-skeleton of $\mathbb{C}P^\infty$. This is a map $S^2 \to \mathbb{C}P^\infty$. Also recall that we have the adjunction $[\Sigma X, Y] \cong [X, \Omega Y]$. Writing $S^2 = \Sigma S^1$ induces a map $S^1 \to \Omega\mathbb{C}P^\infty$. Show that this map is a weak equivalence.

2. Recall that there is a line bundle ξ over $\mathbb{R}P^n = \{\textit{lines in}\, \mathbb{R}^{n+1}\}$ where the fibre over each point is the corresponding line, and an analogous bundle $\xi^\perp$ the fibre over each point is the orthogonal complement of the corresponding line. Show that $T\mathbb{R}P^n \cong Hom(\xi, \xi^\perp)$. What is the corresponding statement for $\mathbb{C}P^n$?

3. Prove that the Eilenberg Maclane spaces $K(\mathbb{Z}, n)$, $K(\mathbb{Z}/2, n)$ cannot be finite dimensional CW complexes for any $n \geq 1$.

4. Compute all Steenrod operations on $H^*(\mathbb{R}P^\infty \times \mathbb{R}P^\infty; \mathbb{Z}/2)$.

5. Compute all Steenrod operations on $H^*(\mathbb{C}P^n/\mathbb{C}P^m; \mathbb{Z}/2)$ for $m < n$.

6. Prove that any complex line bundle over a CW complex is trivial if its restriction to the 2-skeleton is trivial.

7. Prove that $\Omega\mathbb{H}P^n \simeq S^3 \times \Omega S^{4n+3}$.

8. Consider the inclusion $\mathbb{C}P^n \to \mathbb{C}P^\infty$. Prove that the homotopy fibre of the inclusion $\simeq S^{2n+1}$.

9. Let $P(m)$ be the space $S^m \times \mathbb{C}P^1/C_2$ with C_2 acting via

$$\sigma(x, [z_0, z_1]) = (-x, [\bar{z_0}, \bar{z_1}])$$

This is an example of a Dold manifold.
a) Prove that $P(m)$ is orientable if and only if m is even.
b) Find a CW complex structure on $S^m \times S^2$ on which C_2 acts by permuting cells. Using this write down a CW complex structure on $P(m)$ and write down its cellular chain complex.
c) Compute the cohomology of $P(m)$ with $\mathbb{Z}$, $\mathbb{Z}/2$ and $\mathbb{Z}/4$ coefficients. Compute $\beta : H^2(P(m); \mathbb{Z}/2) \to H^3(P(m); \mathbb{Z}/2)$.
d) Show that $S^m \times S^2 \to S^m$ is a C_2-map and hence induces $p : P(m) \to \mathbb{R}P^m$ which is a fibre bundle with fibre S^2. Construct a section $i : \mathbb{R}P^m \to P(m)$. Also observe that there are sections homotopic to i whose image is disjoint from i.
d) Define $v = p^*(x) \in H^1(P(m); \mathbb{Z}/2)$ where $H^1(\mathbb{R}P^m; \mathbb{Z}/2) \cong \mathbb{Z}/2\{x\}$. Let u be the generator of $Ker(i^* : H^2(P(m); \mathbb{Z}/2) \to H^2(\mathbb{R}P^m; \mathbb{Z}/2))$. Prove that $[P(m)] \cap u = i_*[\mathbb{R}P^m]$.
e) Show that $u^2 = 0$. [Hint : Compute the Poincaré dual of u^2.]
f) Prove that $H^*(P(m); \mathbb{Z}/2) \cong \mathbb{Z}/2[v, u]/(v^{m+1}, u^2)$. Also prove that $P(m) \not\simeq S^2 \times \mathbb{R}P^m$.